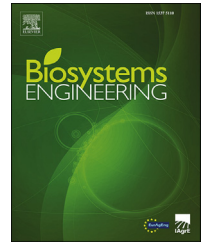


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Research Paper

Spatiotemporal graph convolutional network for automated detection and analysis of social behaviours among pre-weaning piglets



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In the pig industry, social behaviors of preweaning piglets are critical indicators of their livability, growth, health, and welfare status, for which there is an urgent need for using precision livestock farming tools. In this study, a novel method based on graph convolutional networks (GCNs) was developed to characterize preweaning piglet social behaviors such as snout–snout as well as snout–body social nosing and snout–snout as well as snout–body aggressive/playing behavior. Using an integrated CNN-based network, the proposed method first detected and tracked individual piglets. After that, a self-adaptive spatial affinity kernel function was used to detect suspected social behaviors and spatiotemporal graphs with high-quality node features (coordinates, node-estimation confidence, distance from each node to the centroid, and node motion) were built for pairwise piglets for further analysis. The spatiotemporal graph sequences were fed into a self-adaptive GCN combined with an attention mechanism to classify the suspected social behaviours. Our method performed well in detecting piglet social behaviours with a recall of 0.9405, a precision of 0.9669, and an F_1 score of 0.9535. In an 8-h video episode, the time budgets of snout–snout as well as snout–body social nosing and snout–snout as well as snout–body aggressive/playing behaviour were 33.82%, 38.34%, 13.74%, and 14.10%, respectively. The findings show that detecting body part-associated social behaviours in piglets using a GCN is feasible, yielding practical computer vision technologies for improved piglet behaviour monitoring and management.

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Nomenclature

Symbols

bl	The average body lengths of all piglets in a pen
F_1	F_1 -measure
P_n	Precision for node estimation
$Precision_s$	Precision for social behaviour detection
R_n	Recall for node estimation
$Recall_s$	Recall for social behaviour detection
t	Time step

Abbreviations

AP	Average precision
CNN	Convolutional neural network
FP	False positive
FN	False negative
IDS	Identity switch
ID	Identity
IDF ₁	Identity F_1 score
IoU	Intersection over union
LSTM	Long short-term memory
MOT	Multiple object tracking
MOTA	Multiple objects tracking accuracy
OPTN	On-line piglet tracking network
SA	Spatial affinity
TP	True positive
TN	True negative

1. Introduction

With the advancement of deep learning in recent years, computer vision technologies are increasingly being applied in animal science and agriculture for various applications such as monitoring animal behaviours. The pig industry is one of the most prominent domains for such applications (Benjamin & Yik, 2019; Chen, Zhu, & Norton, 2021). In the pig industry, convolutional neural network (CNN), a data-driven feature extraction method in the computer vision community, has been widely used in pig behaviour monitoring, such as the detection of pig feeding behaviours (Alameer et al., 2020; Yang et al., 2018a), identifying pig postures and postural changes (Yang et al., 2021), and recognizing pig tail-biting behaviour (Liu et al., 2020). In general, the CNN-based methods used to perform these detection tasks in pig behaviour monitoring have primarily used video data modality of pig appearance. The modeling of dynamic node-based graphs has rarely been reported in animal behaviour detection studies, which merits further investigation.

The node-based graph lacks a two-dimensional (2D) or three-dimensional (3D) grid, limiting the application of proven models such as CNNs. Recently, graph convolutional networks (GCNs), which generalize CNNs to graphs with arbitrary structures, have gained considerable attention in the community of human behaviour detection and achieved promising results (Liu et al., 2020a; Uluhan et al., 2020; Yan et al., 2018). Graph features have several advantages than

those of RGB images or optical flow because they are concise, powerful, and simple to calibrate and understand (Wu et al., 2019). However, in those human behaviour detection studies (e.g., handshaking), the graphs were constructed for every individual without considering the relationships between individuals (Liu et al., 2020b; Uluhan, Iftekhar, & Manjunath, 2020; Yan et al., 2018). For humans and animals, the relationship between interactive behaviour performers are very critical for detecting social behaviours and should not be ignored. For example, the connection between the snouts of two interacting pigs in a social nosing event should be considered in the graph construction because the majority of pig social behaviours are performed by considerably associating with nosing body parts (Camerlink & Turner, 2013). Furthermore, node features and node connections (edges) are crucial in describing behaviours in graphs. However, in previous studies, only co-ordinates and node-estimation confidence were used as node features (Liu et al., 2020b; Uluhan et al., 2020; Yan et al., 2018). Because these features do not represent motion intensity and node clustering, further research on GCNs with additional node features (e.g., node displacement and distance from the node to centroid) on animal behaviour detection is necessary.

Thus far, automated detection of multiple social behaviours in pigs has not been reported. In the swine industry, the social behaviours of preweaning piglets are critical indicators of their growth, health, and welfare status. Therefore, there is an urgent demand for using precision livestock farming tools (Gómez et al., 2021). Generally, piglet social behaviours are classified into three main categories: social playing, aggressive behaviour, and social nosing (Beirendonck et al., 2014; Verdon et al., 2019). Aggressive behaviour usually results in stress, injuries, and economic losses (Clark & D'Eath, 2013). Pig social playing is very similar to aggressive behaviour (Blackshaw et al., 1997; Camerlink et al., 2016; Held & Špinka, 2011) and potentially indicates favorable environmental conditions for piglets (Held & Špinka, 2011; Lawrence, 1987). Social nosing serves as an affiliative behaviour that accelerates piglet growth (Camerlink et al., 2012; Camerlink & Turner, 2013). Generally, nosing behaviour is performed at a milder intensity than that of aggressive or playing behaviour.

In addition, during a pig social nosing event, snout-to-snout contact tends to be a positive and affiliative nosing behaviour whereas snout-to-tail contact is sometimes regarded as a precursor to harmful behaviour such as tail biting (Camerlink & Turner, 2013). Therefore, distinguishing between snout-to-snout and snout-to-tail social behaviours in pigs and welfare monitoring is critical.

In this study, we used a CNN-based framework to detect piglet social behaviours and analyze the proportions of body part-associated social nosing and aggressive/playing behaviour. Piglets were first detected and tracked using our previous method (Gan et al., 2021a) using the number of piglets to improve tracking performance (IDF1 score of 96.55% and MOTA of 97.04%). Using a CNN node detector, piglet body nodes (i.e., snout, hip, neck, and trunk) were precisely detected after piglet detection. Then, to classify suspected behaviours, a spatiotemporal GCN was used. We conducted experiments to validate the feasibility of classifying piglets

social behaviours based on body parts using the spatiotemporal GCN. We aimed to develop a CNN-based method for automating the detection and analysis of piglet social behaviours, which could potentially lead to a development of a powerful precision livestock farming tool for better pig farm management.

2. Materials

2.1. Animals and video recordings

Data were collected from a pig farm in Foshan City, Guangdong, China. The study was conducted on 10 Meihua sows, a regional heritage pig breed. All sows and their offsprings were individually raised in open floor farrowing pens along (8–12 piglets observed from 2 to 21 postpartum days). The sows had a distinct color pattern with black and white trunk, black head, ears, hip, and tail and white nose tip, belly, and limbs, and almost all the piglets were completely black except a few mottled piglets (see Fig. 1). The farrowing pens, which are approximately $3.8 \text{ m} \times 2.0 \text{ m}$ in size and with a fully concrete floor, allowed the sow and piglets to roam freely throughout the lactation period.

A camera (DS-2CD1321D-I, Hikvision®, Hangzhou, China) was installed above each pen at the height of 2.4–2.7 m, allowing a bird-view recording of the entire pen. The videos were recorded from 7:00 to 18:00 (frame rate: 10 FPS and resolution: 1920×1080 pixels). To reduce the computational cost,

the video frame rate was adjusted to 5 FPS (one every other frame was removed) in the subsequent experiments after data acquisition and the images from the videos were resized to 960×540 pixels using the bilinear interpolation method (Sa, 2015) to reduce computational cost.

2.2. Definition of social behaviours and body key nodes

Detecting the social behaviours of piglets when they are closely gathered and engaged in suckling teats or massaging sow udders are quite complex even via manual observations. According to previous studies (Camerlink & Turner, 2013; Weller et al., 2019), the social behaviours of piglets were mostly exhibited when piglets were not suckling teats or massaging udders. Therefore, piglet social behaviours during nursing were not considered in this study. Similarly, piglet interactions during sow nursing were eliminated by employing the sow nursing behaviour detection method developed and validated in our previous study (Yang et al., 2018). Furthermore, because human observations make it difficult to distinguish between social playing and aggressive behaviour of piglets (Blackshaw et al., 1997; Camerlink et al., 2016; Held & Špinka, 2011), unless otherwise specified, social playing was not distinguished from aggressive behaviour in this study.

To characterize different types of piglet social behaviours such as snout–snout as well as snout–body social nosing and snout–snout as well as snout–body aggressive/playing behaviour, the piglet's snout (e.g., Nodes 1 and 7 in Fig. 2) and hip (e.g., Nodes 4 and 10 in Fig. 2) were identified. Table 1

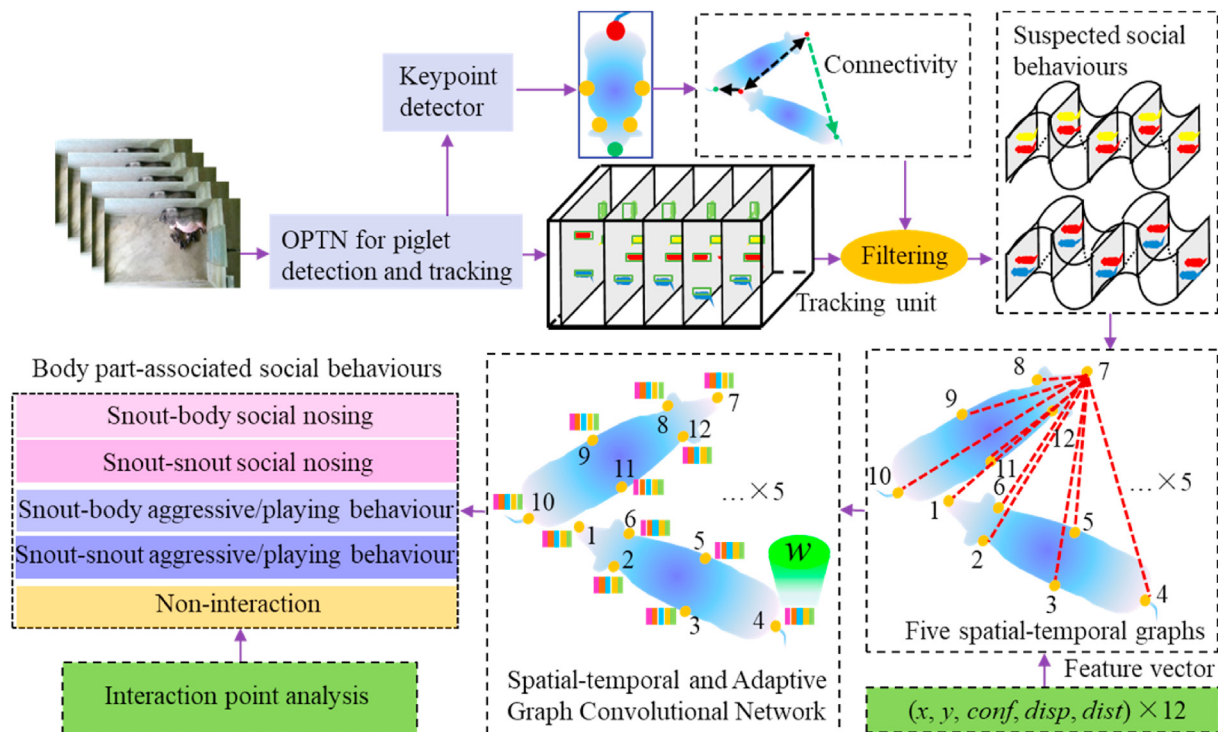


Fig. 1 – Pipeline of detection of piglet social behaviours. ■■■ denotes a five-dimensional feature vector attached to a graph node. w denotes the weights of the STAGCN, x and y denote 2D coordinates in the local pixel coordinate system, conf denotes confidence scores of nodes prediction, disp denotes node displacement between adjacent images, and dist denotes the distance from the node to the centroid.

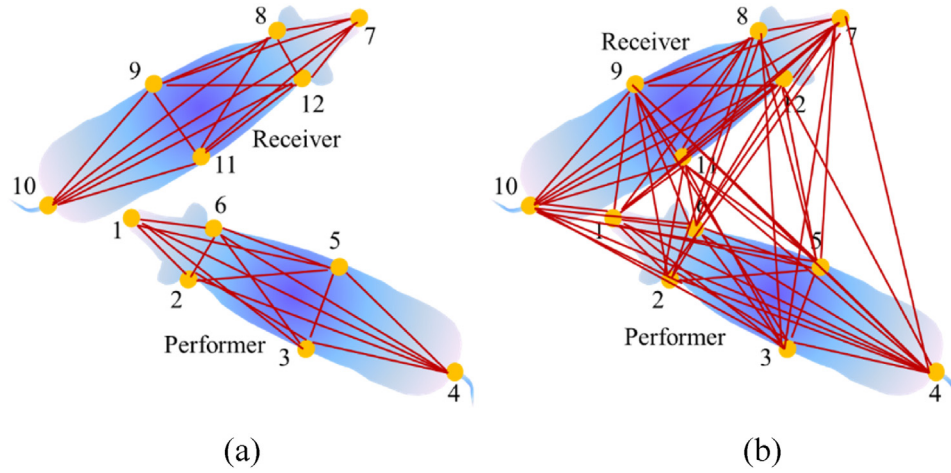


Fig. 2 – Numbered nodes for social behaviour performer and receiver. (a) The partial-connection strategy, (b) The full-connection strategy.

Table 1 – Behaviour descriptions.

Behaviour	Description
Social nosing behaviour	Piglet touching or sniffing any part of the head or nose of another piglet. Snout movements may be observed.
Social playing behaviour	Nudging or pushing, playing, and fighting.
Aggressive behaviour	Forceful fighting, pushing with the head, or violently biting littermates.

contains detailed descriptions of all key nodes and social behaviours. For example, suppose the snout of a social behaviour performer is closer to the snout of a receiver in any twointeracting piglets. In that case, the social behaviour is considered snout–snout, e.g., snout–snout social nosing and playing/aggressive behaviour. Otherwise, it will be classified as a snout-body social behaviour, i.e., snout-body social nosing and playing/aggressive behaviour (Camerlink et al., 2016).

2.3. Data labeling

The video data was divided into two datasets: training and test. To begin, the 10 experimental pens were randomly shuffled and the data from the first seven pens were employed as a training dataset for piglet detection and tracking as well as and keypoint (node) detection. The remaining three videos were used as a test dataset to evaluate the results of piglet detection, node estimation, and piglet tracking. Each pen's videos were then trimmed to 30-s video clips, and 10 clips were chosen from each pen. In these selected video clips, piglets were exhibiting aggressive/playing behaviour, nosing, chasing, hustling, and sleeping. In addition, adhesion between piglets and occlusions from the sow was observed. In addition, an 8-h video episode from one of the three test pens was reserved for further validation. During the 8-h video episode, piglets were actively exhibiting social behaviours during the day and the sow occasionally performed nursing behaviour throughout the duration. As a result, the study used 100 video clips (70 for training and 30 for test) and one long video episode.

Animal behaviorists guided the labeling of real-life piglet social behaviours in the video clips. Each piglet's bounding box and nodes were manually labeled first with the software *Labelme* (<https://github.com/wkentaro/labelme.git>). Each piglet's ID number in each pen was annotated, beginning with “1” and ending with the total number of piglets in the pen. The images were used to identify pairs of piglets which had physical contact. These pairs of piglets were classified into three categories based on the ethogram in Table 1: non-interaction, social nosing, and aggressive/playing behaviour. The video names, IDs of pairwise piglets, and timing of the social behaviours (video sequence number) were all recorded. In our dataset, any pair of piglets exhibiting social behaviour in 5 successive frames (1 s) was considered a positive sample (social nosing, or aggressive/playing behaviour) and a 5-frame video clip could contain multiple social behaviour events.

3. Methods

3.1. Overview of the online tracking method

To utilize the power of CNNs, we developed a deep learning network system primarily comprising three CNNs: a completely integrated CNN-based online piglet tracking network (OPTN) for piglet detection and tracking (Gan et al., 2021a), a keypoint detector, deep high-resolution network (DHRN) (Sun et al., 2019), and a suspected behaviour classifier. The pipeline of the proposed method is depicted in Fig. 1. Firstly, piglets in video clips were specifically detected and tracked using OPTN. Upon piglet detection, a CNN-based node detector was used to identify six

piglet body nodes, including the snout, hip, neck, and trunk. The spatial relationship between pairwise piglets was represented a self-adaptive spatial affinity kernel function. Next, suspected social behaviours of piglets in a tracking unit (Sect.3.2.1) were detected using spatial affinity estimations. Following that, an interactive graph was created for each pair of piglets that might be engaging in social behaviour. Finally, a spatiotemporal and adaptive GCN (STAGCN) was used with a sequence of 12-node spatiotemporal graphs for a suspected social behaviour to further classify the suspected social behaviour as social nosing, aggressive/playing behaviour, and noninteraction behaviour. As an example, the connectivity between Node 7 and all other nodes is shown in Fig. 1. In fact, there were numerous permutations of the two-node connectivity.

3.2. Detection of suspected social behaviour

3.2.1. Piglet tracking and node detection

To extract the spatiotemporal features of pairwise piglet social behaviours, we used OPTN to detect and track piglets, which was developed and validated in our previous study. The OPTN can detect and track piglets simultaneously and is robust against occlusion and dim lighting because it uses a distance-based strategy that can recover missing or occluded identities. To obtain a stable tracking trajectories of pairwise piglets, video clips were equally segmented into a quantity of 1-s clips, each containing five successive video frames because a social behaviour event will only be classified if the piglets engaged in that behaviour for at least 1 s (Liu et al., 2020a; Oczak et al., 2014). We strictly followed the ethogram in Table 1 when determining the duration of a social behavior event for every frame of a 1-s clip, particularly the first and last frames. Moreover, we referred to the video clips that came before and after the current one for each video clip, and at least two people scored each sample to ensure accuracy. Each 1-s clip was referred to as a tracking unit in this case, and each tracking unit was processed independently. Each tracking unit's trajectory began at the first frame and ended at the last frame. DHRN, a multi-scale convolutional network, was used to detect piglet key nodes (Fig. 2) for graph construction, and a GCN was used to classify piglet social behaviours. Sect. 3.3.1 explains the node selection procedure.

3.2.2. Spatial affinity evaluation

The snout was the main body part engaged in performing social behaviours by piglets and was close to the behaviour receiver in social interaction, according to behaviour characteristics of piglets (Camerlink & Turner, 2013). To manifest the spatial relationship between two piglets, a self-adaptive kernel function was developed to model the spatial affinity between the two pairwise piglets and only the piglets engaging in suspected social behaviours within a tracking unit were retained for further analysis to reduce the computation cost. The spatial affinity value was denoted as SA, which can be computed as follows:

$$SA = 3 \times bl \left/ \min \left(\sum_{t=1,3,5} d_{12}^t, \sum_{t=1,3,5} d_{21}^t \right) \right. \quad (1)$$

where d_{12}^t is the sum of Euclidean distances from the snout of piglet₁ to the snout and hip of piglet₂ at time step t , d_{21}^t is the

sum of Euclidean distances from the snout of piglet₂ to the snout and hip of piglet₁ at time step t , and bl was the average body length of all piglets in a pen. To reduce the impact of any missed detection, only the 1st, 3rd, and 5th frames were chosen for computing SA. If $d_{12}^t < d_{21}^t$, piglet₁ was regarded as the performer of the behaviour and piglet₂ as the receiver, and vice versa.

3.3. Classification of suspected social behaviours using a GCN

3.3.1. Graph construction

Node. The first step in creating interactive graphs is to identify nodes that are closely related to social behaviour identification. In this study, six nodes were chosen on each piglet (one on the snout, one on the hip, two on the neck, and two on the trunk, yielding 12 nodes on each interactive graph. These nodes were numbered anticlockwise for the performer and receiver, respectively (Fig. 2). The following are the reasons for choosing these six nodes: Snout was chosen as one of the candidate nodes (Nodes 1 and 7) owing to the snout-centric trait of piglet social behaviour. Given the behaviour dynamics of piglets, the two neck nodes were chosen candidate nodes to improve the distinction between aggressive/playing behaviour and social nosing (Nodes 2, 6, 8, and 12). For example, piglets shake their heads more violently in aggressive/playing behaviour than in social nosing. The two trunk nodes (Nodes 3, 5, 9, and 11) were chosen because trunks were crucial in behavioural expression, particularly when piglets bent their bodies. Regarding the hips of piglets (Nodes 4, 10), the orientation from hip to snout could indicate the body direction and the change in orientation could indicate body motion such as rotation.

Edge. Interactions between piglets were either preceded or followed by considerable associations between the performer and receiver of a social behaviour (Fig. 2) without any direct physical connectivity. An interactive graph was constructed with an exhaustive permutation of connectivities (edges) between two nodes to establish the relationship between a pair of social behaviour performer and receivers, which was referred to as the full-connection strategy (Fig. 2b). However, the partial-connection strategy is a method in which connections (edges) are established only within a piglet (Fig. 2a). The strategies employed an adjacent matrix A elemental values of 1 and 0 representing “with connection” and “without connection”, respectively.

Instead of using a fixed adjacent matrix A to represent the physical connection (edge) between nodes, to make our method more adaptive to the data, we used a learnt adjacency matrix M in our study to improve the features of key nodes and edges that dominate the behaviour expression at each time step. Similarly, the graph convolution operation performed on a node can be given as follows:

$$f_{out} = \sum_k^{K_n} W_k f_{in} (A + M(f_{in})) \quad (2)$$

where f denoted the feature map and W denoted the weighting function, which was similar to the original convolution operation, in that it comprised a set of weight vectors with the

same dimension as the input features. M determines the edge strength and varies with the input feature map, which may facilitate the STAGCN to learn more discriminative features in a data-driven fashion.

3.3.2. Node features

In addition to the conventional node features such as coordinates (x_{loc} , y_{loc}) and node-estimation confidence ($conf$), we used distance ($dist$) and displacement ($disp$) as node features to enhance the spatial and temporal graph features, respectively. The $disp$ represents the node displacement between adjacent frames, which featured the inter-frame edges for one specific node i and represented its trajectory over time, indicating the intensity of the social behaviour. The $dist$ was the distance between a node and the centroid of the interactive graph, which was an intra-frame feature that indicated the cluster of nodes. As illustrated in Fig. 3, to demonstrate the definitions of the features, an aggressive/playing case is used as an example. We constructed an undirected spatiotemporal graph $G = (N; F)$ with 5 frames and 12 joints on each frame, with intra-frame and inter-frame connections in the tracking unit. In G , the node set $N = \{n(x_{t,i}, y_{t,i}) | t = 1, 2, 3, 4, 5, i = 1, 2, 3, \dots, 12\}$, where $n(x_{t,i}, y_{t,i})$ denotes the coordinates of a node in source image pixel coordinate system and

$$x_{\max} = \max(\{x_{t,i} | t = 1, 2, 3, 4, 5, i = 1, 2, 3, \dots, 12\}) \quad (3)$$

$$x_{\min} = \min(\{x_{t,i} | t = 1, 2, 3, 4, 5, i = 1, 2, 3, \dots, 12\}) \quad (4)$$

$$y_{\max} = \max(\{y_{t,i} | t = 1, 2, 3, 4, 5, i = 1, 2, 3, \dots, 12\}) \quad (5)$$

$$y_{\min} = \min(\{y_{t,i} | t = 1, 2, 3, 4, 5, i = 1, 2, 3, \dots, 12\}) \quad (6)$$

Then the coordinate features (x_{loc} , y_{loc}) can be calculated by normalizing $n(x_{t,i}, y_{t,i})$ into a local coordinate system derived from $x_{\max}$, $x_{\min}$, $y_{\max}$, and $y_{\min}$:

$$x_{loc} = \frac{x_{t,i} - x_{\min}}{x_{\max} - x_{\min}} \quad (7)$$

$$y_{loc} = \frac{y_{t,i} - y_{\min}}{y_{\max} - y_{\min}} \quad (8)$$

The $disp$ and $dist$ were calculated in the local coordinate system.

As an input to the STAGCN, the feature vector on a node $F(x_{t,i}, y_{t,i})$ will be (x_{loc} , y_{loc} , $conf$, $dist$, and $disp$), of the i -th joint on frame t .

3.3.3. Graph convolution

Partitioning strategy. Because graphs are in the form of 2D or 3D grids, node partitioning is required for CNNs to be utilized in graph convolution. A spatial configuration partitioning strategy (Yan et al., 2018) was used to map each node with a unique weight vector (Fig. 4). The kernel size was empirically determined to be 3, and all nodes were divided into three subsets: root nodes (yellow), centripetal nodes (blue), and centrifugal nodes (red). All nodes shared the weight vector in the same subset.

Network. According to the attention mechanism proposed by Shi et al., (2020), the spatial, temporal, and channel

attention mechanisms (STC-attention, Fig. 5) were adopted in our study to improve the quality of extracted features from spatial, temporal, and channel aspects, respectively. The three attention mechanisms were coded sequentially with different orders since Shi et al., (2020) proved that the sequential order outperformed the parallel one.

A basic block was formed by one spatial GCN, one temporal GCN, and a shortcut (Fig. 5). The partitioning strategy used resulted in the same convolution for the temporal dimension Yan et al., (2018). Following the spatial and temporal GCNs, a batch normalization layer and a ReLU layer were used. To stabilize the training, a residual connection (He et al., 2016) was added to each basic block, allowing the layer to be inserted into any existing model without disrupting its initial behaviour. If the number of input channels differed from the number of output channels, a 1×1 convolution (temporal conv in the dotted line box) was inserted in the residual path to transform the input to match the output in the channel dimension. Our STAGCN has ten basic blocks and a total of [32, 32, 32, 32, 64, 64, 64, 128, 128, and 128] output channels. The global average pooling layer and the softmax classifier were used to predict the behaviour categories based on the extracted features.

3.4. Implementation details and comparative experiments

All CNNs were trained and tested using the Pytorch framework and a single NVIDIA GeForce RTX 2080 TI graphics processor. STAGCN's total epochs and initial learning rate were 160 and 10^{-3} , respectively during the training phase. After the 120th epoch, the learning rate decay factor was 10^{-1} . Data augmentation methods (horizontal, vertical flipping, and 180° rotation) were used to improve data generalization. Furthermore, we compared our method to another one developed in a similar study (Liu et al., 2020a). Pig detection, tracking, and behaviour classification were used in both our method and Liu's method. In Liu's study the following observation were made: 1) analogous to our study, pigs were free to roam without occlusion from the environment and videos were segmented into 1-s video clip and 2) unlike our method, SSD was used as a pig detector, geometrical features (e.g., aspect ratio and intersection over unit) were applied for pig tracking, and a CNN combined with long short-term memory (LSTM) was used as a classifier to detect pig tail-biting behaviour.

One of the primary goals of our research is to validate the performance of behaviour classification based on GCN and the detection as well as tracking method in our method (OPTN), which has rarely been validated in a previous study (Gan et al., 2021a). Therefore, for a fair comparison, the dataset and pig detection and tracking methods in our method and Liu's were identical and only the behaviour classification section was compared. In particular, GCN and CNN + LSTM networks were used as a classifier in our method and Liu's method, respectively.

In a previous study (Gan et al., 2021b), we used handcrafted key point-based features to describe the angular and linear motion of pairwise piglets and SVM to classify piglet social behaviours in a dataset similar to the one used in this study. As a result, the current study's method was compared to the previous one, which used the same dataset and piglet detection as well as tracking method (OPTN).

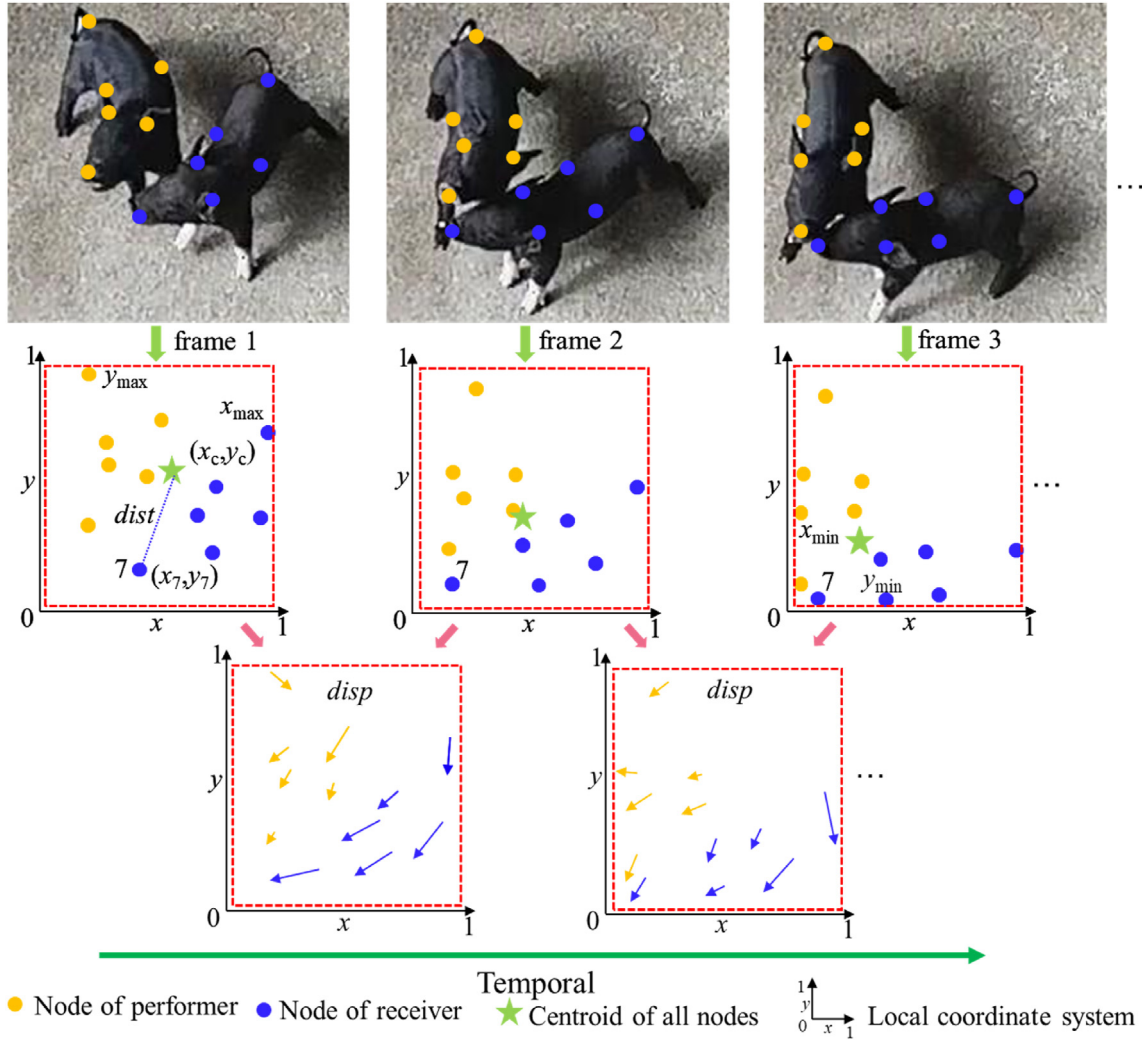


Fig. 3 – Nodes of aggressive/playing behaviour in three successive frames of images. (x_7, y_7) and (x_c, y_c) represent the coordinates of Node 7 and the centroid in a local coordinate system. Only three frames have been shown for brevity. The last row displays the node-level displacement calculated from adjacent video frames.

3.5. Evaluation methods

Average precision (AP_{pig}) was used to evaluate the performance of piglet detection which can be given as follows (Liu et al., 2018):

$$Precision_{pig} = \frac{TP_{pig}}{TP_{pig} + FP_{pig}}, \quad (9)$$

$$Recall_{pig} = \frac{TP_{pig}}{TP_{pig} + FN_{pig}}, \quad (10)$$

$$AP_{pig, \eta} = \frac{1}{101} \cdot \sum_{Recall_{pig}} Precision_{pig}(Recall_{pig}, \eta), \quad (11)$$

where TP_{pig} was the number of detections with IoUs (intersection over union) greater than or equal to 0.5 over ground truth bounding boxes, FP_{pig} was less than 0.5, FN_{pig} was the number of missing detections, $Recall_{pig}$ and $Precision_{pig}$ were calculated using the IoU threshold of 0.5, and $Precision_{pig}$ was averaged (AP) by the number of $recall_{pig} \in [0:0.1:1]$.

CLEAR MOT metrics (Bernardin & Stiefelbogen, 2008) were commonly used in evaluating multiple object tracking and were employed to evaluate the OPTN tracking performance. In addition to CLEAR MOT metrics, IDF_1 score (Ristani et al., 2016), another representative MOT evaluation metric, was used in our study. Two most important metrics MOTA and IDF_1 score were calculated as follows:

$$MOTA = 1 - \frac{\sum (FP_{pig} + FN_{pig} + IDS)}{\sum (GT)}, \quad (12)$$

$$IDF_1 = \frac{2IDTP}{2IDTP + IDFP + IDFN}, \quad (13)$$

where IDS denoted the number of ID switches between adjacent frames.

For the node detector, precision P_n and recall R_n were used to evaluate node detection performance (Girdhar et al., 2018; Hosang et al., 2016; Lin et al., 2014; Ronchi & Perona, 2017). A detected node with an Euclidean distance of less than 10% of D from the corresponding ground-truth node

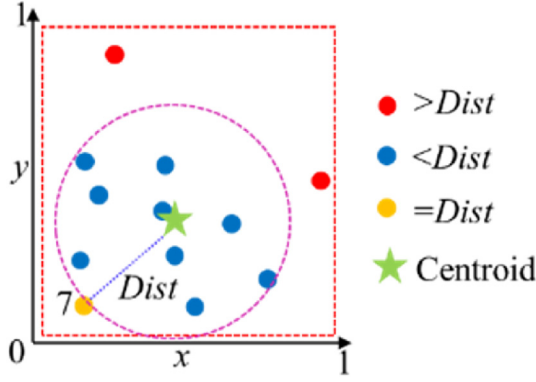


Fig. 4 – Partitioning strategy for constructing convolution operations. The nodes are labeled based on their distances from the centroid (green star) relative to the root node (yellow). Centripetal nodes have shorter distances (blue) than the root node (red). *Dist* is the average distance between a tracking unit's root node and centroid over all frames. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

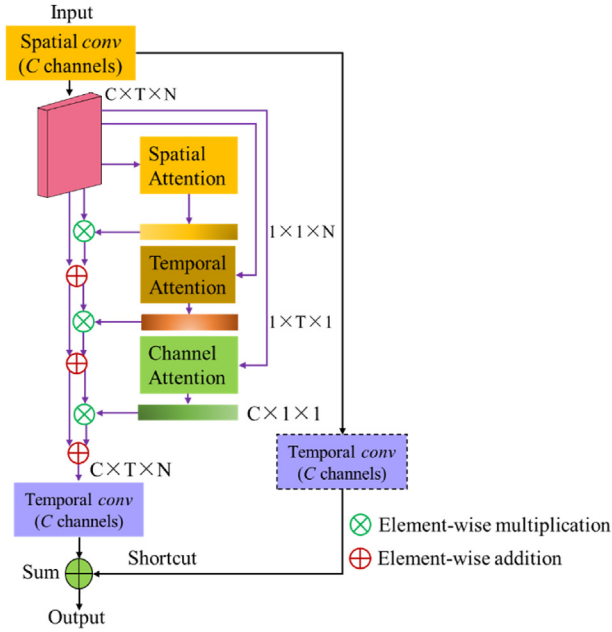


Fig. 5 – The basic block consists of spatial–temporal residual structure and STC-attention. Temporal conv in dotted line box is inserted only when the channel dimensions of input and output features are not equal. Spatial, temporal, and channel attention mechanisms are used between the spatial and temporal convolutional layers.

was considered correctly detected, where D denoted the piglet body length from hip to snout (Andriluka et al., 2014). The values TP_n , FP_n , and FN_n represent the number of nodes that were correctly detected, falsely detected, and missed, respectively.

$$P_n = \frac{TP_n}{TP_n + FP_n} \quad (14)$$

$$R_n = \frac{TP_n}{TP_n + FN_n} \quad (15)$$

As for social behaviour detection, $Precision_s$, $Recall_s$, and F_1 (F_1 -measure) (Hripcsak & Rothschild, 2005) were adopted to evaluate the performance of the proposed method:

$$Precision_s = \frac{TP_s}{TP_s + FP_s} \quad (16)$$

$$Recall_s = \frac{TP_s}{TP_s + FN_s} \quad (17)$$

$$F_1 = \frac{2Recall_s \cdot Precision_s}{Recall_s + Precision_s} \quad (18)$$

where TP_s , FP_s , and FN_s denoted the number of correctly detected, falsely detected, and missed social behaviour events, respectively (Lee et al., 2016). TN_{non} denotes the number of correctly detected noninteractions.

4. Results

4.1. Performance of piglet detection, tracking and node detector

Piglets were detected with an $AP_{pig,0.5}$ of 0.978 and a $Recall_{pig}$ of 0.987 using OPTN, and the were tracked with an IDF1 score of 0.9655 and a MOTA of 0.9704. Piglet nodes were detected with a total recall of 0.9375 and a total precision of 0.9681.

4.2. Performance of detection of social behaviour

The probability distributions of the spatial affinity of contact and noncontact pairwise piglets from training dataset are depicted in Fig. 6. The statistical analysis revealed that the probability density overlaps between contact and noncontact pairwise piglets were approximately [0.7, 0.9]. As a result, a spatial affinity threshold of 0.7 was set to include more suspected social behaviours while filtering out most noncontact cases.

4.2.1. Evaluation of our method in short video clips

The contributions of various feature components (*conf*, *disp*, *dist*, *x*, and *y*) in our framework were investigated by disabling one feature component at a time and examining the performance drop in terms of F_1 on the test dataset. A full-connection strategy's detection performance was compared to a partial-connection strategy. As illustrated in Fig. 7, the full-connection strategy combined with all features had the highest F_1 score of 96.76%. Thus the full-connection strategy combined with all features was further applied in the subsequent experiments.

With a recall of 0.9405, a precision of 0.9669, an F_1 score of 0.9535, and a TN_{non} of 867, the STAGCN successfully classified suspected social behaviours (Table 2). Social nosing had a recall of 0.9543, a precision of 0.9664, and an F_1 score of 0.9602.

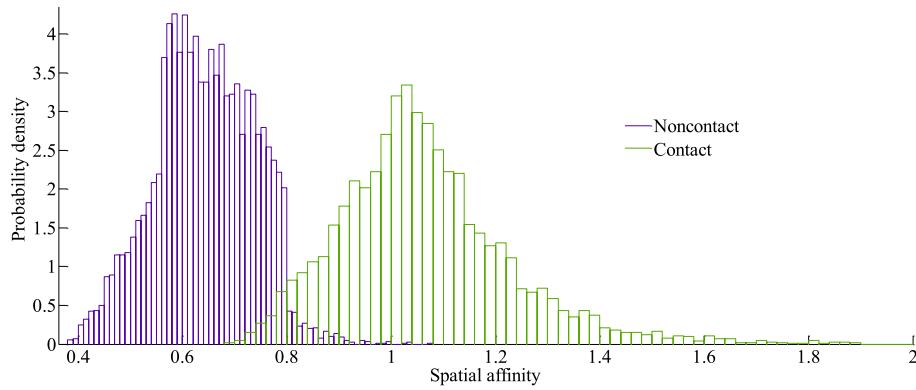


Fig. 6 – Spatial affinity between contact piglets (green) and noncontact piglets (purple). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Aggressive/playing behaviour was detected with a recall of 0.9118, a precision of 0.9681, and an F_1 score of 0.9391. The inference speed was 6.35 fps. In comparison to the study by Liu et al., (2020a), the results revealed that STAGCN performed better in social behaviour classification with recall, precision, and F_1 score of 4.61%, 9.37%, and 6.98% higher, respectively. Furthermore, our model was 26.47% smaller in size and has a faster inference speed (by 8.61%). When compared to the method used in a previous study (Gan et al., 2021b), the results showed that STAGCN performed better in classification with $Precision_s$ (+4.02%) and F_1 (+1.49%) but with lower $Recall_s$ (−1.03%).

To further analyze the performance of our method, a confusion matrix (Table 3) is provided. Finally, examples of social behaviour detection under various conditions are shown in Fig. 8.

4.2.2. Graph topology visualization

We visualized the edge strength (the learnt adjacency matrix) in various behaviours from different residual basic blocks (the 1st, 3rd, 5th, 7th, and 9th block) and displayed two representative examples from playing behaviour and social nosing (Fig. 9), respectively. Similarly, we visualized the learned spatial attention in different behaviours from different residual basic blocks in Fig. 10.

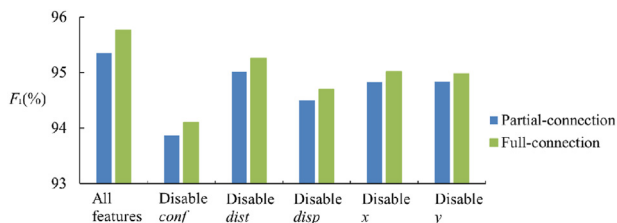


Fig. 7 – Comparative performance (F_1) by disabling feature components (conf, disp, dist, x, and y), respectively and using all features. Detection performances of full-connection strategy and partial-connection strategy are compared.

4.2.3. Continuous data analysis in a long video episode

Our method was further tested in an 8-h long video episode recorded during the day. The time budgets for snout–snout and snout–body social nosing were 33.82% and 38.34%, respectively, and that of snout–snout and snout–body aggressive/playing behaviour were 13.74% and 14.10%, respectively (Fig. 11). The number of tracking units of social nosing or aggressive/playing behaviour was totaled per 15-min session (Ott et al., 2014). As shown in Fig. 12, the detection results and ground-truth data were compared. Table 4 summarizes our method's performance in detecting social behaviour. Piglet social behaviours were successfully detected with a recall of 0.9221, a precision of 0.9421, and an F_1 of 0.9320. The total number of TN_{non} was 247. Furthermore, 72.16% of social behaviours are attributed to social nosing and 27.84% are attributed to aggressive/playing behaviour.

5. Discussion

To the best of our knowledge, this is the first to use a GCN to classify animal behaviours. Different types of social behaviours (snout–snout as well as snout–body social nosing; snout–snout as well as snout–body aggressive/playing behaviour) associated with different body parts were identified, which is useful in piglet behaviour analysis. Determining the social involving three or more performers based on the ID relationship between pairwise piglets is possible (Fig. 8 b). We discovered the following while working on this project:

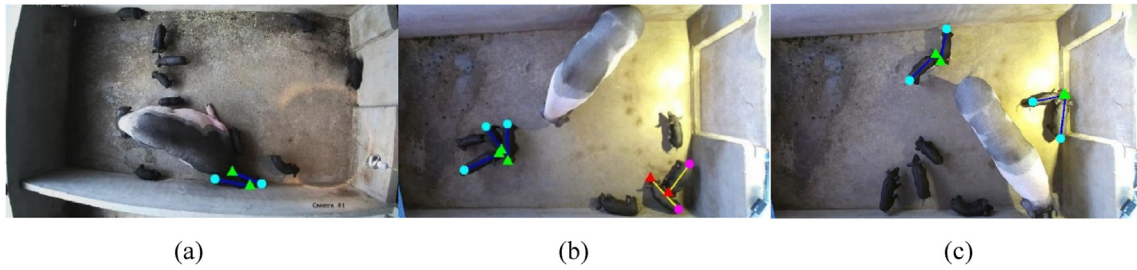
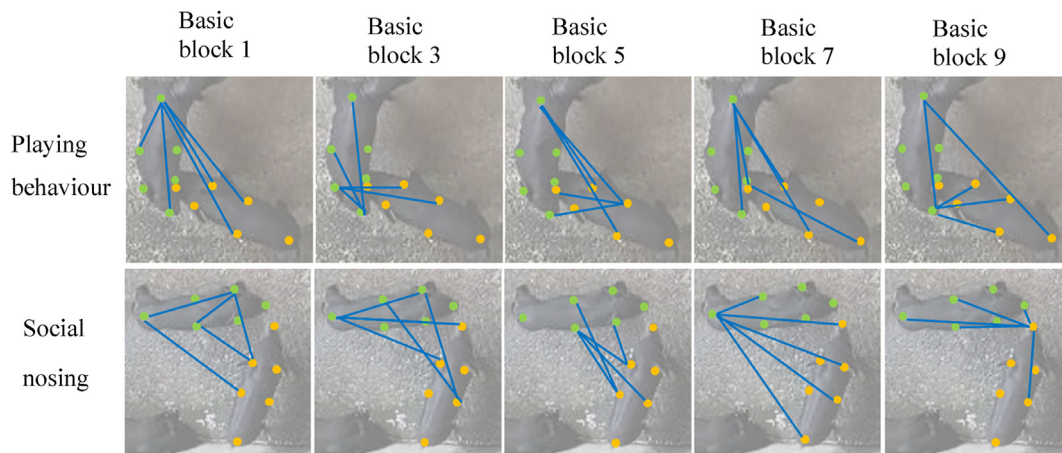
- (1) **High-quality graph construction.** In addition to node-estimation confidence and x and y coordinates, node displacement (disp) and distance (dist) were used as node features in our study to describe the graph. The GCN achieved the best classification performance when all five features were used as graph features (Fig. 7). This suggests that node motion information is critical for graph-based behaviour recognition. Every component in Fig. 7 benefited from the full-connection strategy in graph construction, demonstrating the efficacy of the full-connection strategy. The spatiotemporal graphs were extracted from a self-adaptive GCN that was

Table 2 – Detection and classification results of social behaviours.

Method	Behaviours	TP _s	FN _s	FP _s	TN _{non}	Recall _s	Precision _s	F ₁	Speed /fps	Model size/MB
Ours	Social nosing	1064	51	37		0.9543	0.9664	0.9602	6.31	
	Aggressive/playing	486	47	16	887	0.9118	0.9681	0.9391		575
	Total	1550	98	53		0.9405	0.9669	0.9535		
Liu et al., (2020a)	Social nosing	1025	90	157		0.9193	0.8672	0.8925	5.81	
	Aggressive/playing	449	84	57	859	0.8430	0.8874	0.8646		782
	Total	1474	174	214		0.8944	0.8732	0.8837		
Gan et al. (2021b)	Social nosing	1076	39	103		0.9650	0.9126	0.9369		
	Aggressive/playing	491	42	21	816	0.9212	0.9589	0.9388	6.35	571
	Total	1567	81	124		0.9508	0.9267	0.9386		

Table 3 – Confusion matrix for classifying suspected piglet social nosing behaviours.

		Classified behaviours		
		Noninteraction	Social nosing	Playing/aggressive behaviour
Labeled behaviours	Noninteraction	887	1	0
	Social nosing	0	1064	16
	Playing/aggressive behaviour	0	36	486

**Fig. 8 – Social behaviour detection results under difficult conditions. (a) Detection performance under low lighting conditions. (b) Detection of multiple social behaviour performers. (c) Detection performance under heat lamp. $\blacktriangle$ – $\bullet$ denotes social nosing and $\blacktriangle$ – $\bullet$ denotes aggressive/playing behaviour.****Fig. 9 – Visualization of the edge strength in different behaviours. Only the top5 strength edges are given.**

optimized in a data-driven manner, which improved the method's transferability and applicability in different animals and behaviours such as interactive behaviours between piglets and sows, cattle lamming, mating and ruminant behaviours.

We have created well-generalized datasets with various piglet postures and built high-quality graphs to classify the social behaviours. The positive detection results indicated that our method's the data-driven approach and well-generalized datasets enabled strong robustness against

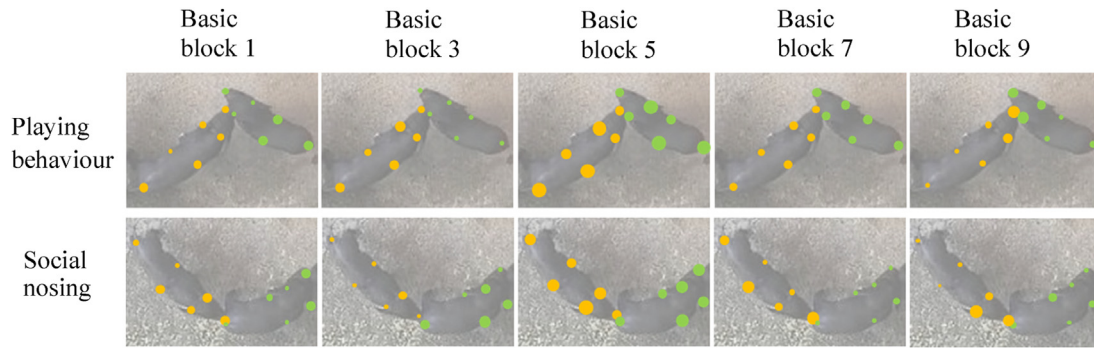


Fig. 10 – Visualization of learned spatial attention in different behaviours. The size of the circle represents the importance of the corresponding node.

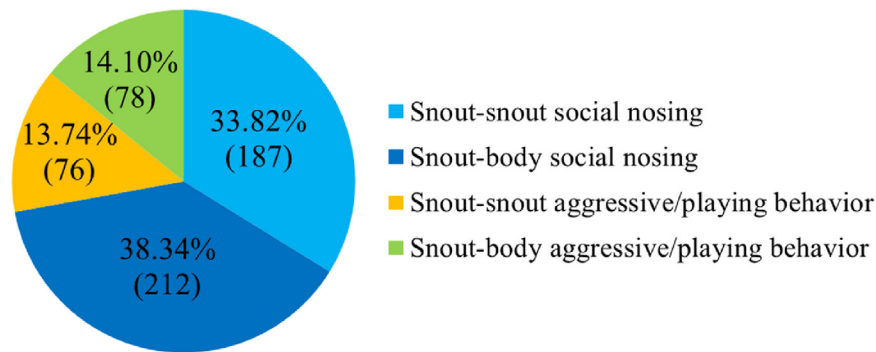


Fig. 11 – Time budget of different types of social behaviours in the long video episode. Numbers in the brackets denote the number of different types of social behaviours.

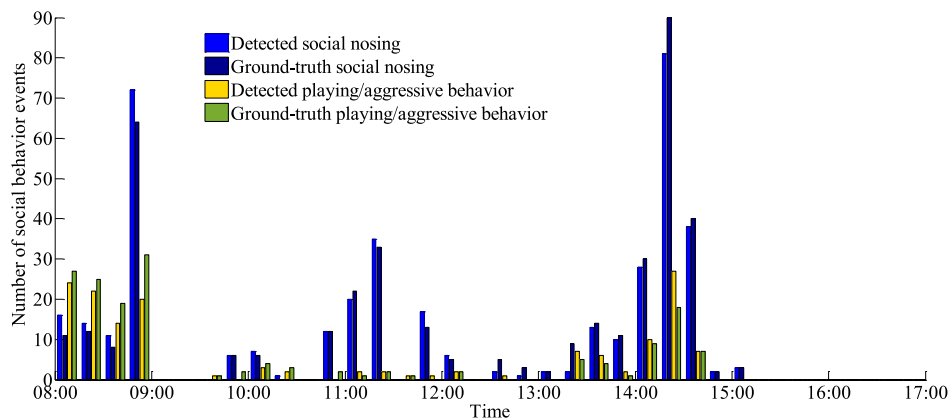


Fig. 12 – The detection results and ground-truth data. The number of tracking units of social nosing or aggressive/playing behaviour was totaled per 15-min session.

Table 4 – Detection and classification results of social behaviours.

Social behaviours	TP _s	FN _s	FP _s	Recall _s	Precision _s	F ₁
Social nosing	373	28	26	0.9302	0.9348	0.9325
Aggressive/playing behaviour	148	16	6	0.9024	0.9610	0.9308
Total	521	44	32	0.9221	0.9421	0.9320

various piglet postures even when the SA values may be close. As a result, piglet postures only marginally affected classifying a suspected social behaviour. For example, pairwise piglets exhibiting sleeping behaviour (in lying posture) or nosing behaviour (in standing posture) could have close SA values but could be classified as noninteraction and social nosing behaviour, respectively using GCN.

- (2) **Graph analysis.** As we used a learnt adjacency matrix, the STAGCN may change the edge importance based on the input feature. Figure 9 demonstrates that connections between two piglets are more important than those within a piglet. This confirms the effectiveness and necessity of allowing a pair of piglets performing a social behaviour to share one graph to establish the two piglets' relationship. Figure 10 indicates that a higher layer contains higher-level information and thus, the graph is more relevant to the final classification task. This could be due to the snout-centric characteristic of piglet social behaviours.
- (3) **Detection result analysis.** We analyzed the piglet social behaviour frequency that varied with time in Fig. 12. We discovered that piglets primarily engaged in social behaviour in the morning (8:00–9:00), at noon (11:00–12:30), and in the afternoon (14:00–15:00). We watched the video again and discovered that piglets dispersed when the sow changed posture from lying to standing, turned actively running around and interacted more with her littermates (Beirendonck et al., 2014). In comparison to Liu's method in Table 2, our method achieved higher recall, precision, and F_1 score of 4.61%, 9.37%, and 6.98%, respectively. We believe that using CNN-based appearance features directly is insufficient for distinguishing piglet social behaviours. Furthermore, because Liu's method is not body part-associated, it cannot perform additional analysis of piglet social behaviour classification. Compared to our previous study (Gan et al., 2021b), the current method has 4.02% higher Precision_s, and 1.49% higher F_1 score but 1.03% lower Recall_s. The comparison results show that the data-driven method STAGCN outperforms the handcrafted feature-based method SVM.
- (4) **Error analysis.** From Tables 2 and 4, we could find that the recall of aggressive/playing behaviours is 4.25% and 2.78% lower than that of social nosing. This is due to the varying levels of difficulty in piglet detection and tracking. Piglets engaged in aggressive/playing behaviours tend to move their bodies more quickly and frequently, which can cause a variety of image problems such as motion blur. These issues can negatively impact object detection and tracking performance. However, piglets engaging in social nosing are moving with a lower intensity indicating that they have fewer difficulties with in object detection and tracking. According to the confusion matrix in Table 3, noninteraction is not the leading cause of false recognition because noninteraction usually refers to sleeping piglets. Sleeping piglets are in a static and motionless state, making it easy for the STAGCN to distinguish them from real social behaviours.

6. Conclusions

We developed a CNN framework combined with a spatio-temporal GCN-based classifier to detect social behaviours in pre-weaning piglets. As a result, our method achieved good

detection performance in short video clips, with a recall of 0.9405, precision of 0.9669, and F_1 score of 0.9535. For example, in an 8-h video clip, the time budgets for snout–snout and snout–body social nosing as well as snout–snout and snout–body aggressive/playing behaviour were 33.82%, 38.34%, 13.74%, and 14.10%. The positive detection performance demonstrated the feasibility of detecting body part-associated social behaviour in piglets using GCN.

Animal ethical welfare statement

All of the animal experiments in this study were conducted in compliance with a protocol reviewed and approved by the Institutional Animal Care and Use Committee in South China Agricultural University.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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